

THE ZERO PARADOX

ZP-K: Computational Grounding of Self-Reference

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This document establishes the computational grounding of the Zero Paradox's central self-reference structure. On the framework's reading, $\perp$ in the computational instantiation is not a state of a Turing machine: it is the ground state of a universal Turing machine — the state from which no external executor is required. Kleene's second recursion theorem supplies a fixed point of the self-application operator; that this fixed point is to be read as the computational expression of $\perp = \{\perp\}$ is the framework's commitment, carried by the KleeneStructure class, not a property the theorem establishes.

The central result is a three-way equivalence within this framework, in the precise sense defined by Remark R-K.0: the structural roles of $\perp$ — Quine atom (set-theoretic), bottom element (order-theoretic), and join identity (algebraic) — are PROVED to coincide, while the fourth role, Kleene fixed point (computational), is required as a typeclass field rather than derived. (See Remark R-K.0 for the precise scope: equivalences (1)–(3) are derived via T-EXEC; equivalence (4) is combined by KleeneStructure's typeclass requirement, not derived independently.)

Section I: The Kleene Fixed Point

I. Kleene's Second Recursion Theorem

Kleene's second recursion theorem is the computational fixed-point theorem. In the form Mathlib states it, for any partially computable F sending each program code to a partial function, some code c computes exactly $F(c)$ (kleene_fixed_point_exists). Rogers' fixed-point theorem is the form for a total computable transformation g of codes: some code c computes the same function as $g(c)$ (roger_fixed_point_exists). Applied to the self-application map of II below, the recursion theorem gives a program whose output at any input n equals its output at its own Gödel number plus n — a periodicity fixed point.

In Lean 4, this is formalized in Mathlib's computability library. For any partially computable transformation f of codes, there exists a code c such that $\text{eval } c = f c$. The theorem supplies a code without distinguishing one. Section IV tabulates the measured axiom footprints.

Kleene's Second Recursion Theorem (Mathlib)

For any partially computable $f : \text{Code} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$, there exists $c : \text{Code}$ such that $\text{eval } c = f c$.

Applied to the selfApply transformation, this yields a code c satisfying $\text{eval } c n = \text{eval } c (\text{encode } c + n)$ for all n — a periodicity condition with period c 's own Gödel number, though not necessarily the least. For every such f the fixed-point codes form an infinite set (fixed_points_infinite, Kleene.lean § VIII).

II. The Self-Application Map

The self-application map sends each code c to the partial function that runs c on c 's own Gödel number plus an offset. A fixed point of self-application satisfies a periodicity condition: $\text{eval } c \ n = \text{eval } c \ (\text{encode } c + n)$ for all n , with period equal to c 's own Gödel number, though not necessarily the least. The periodicity condition is met by codes that ignore their input: every constant code satisfies it, and `infinite_quine_family` witnesses an unbounded family of fixed points with exactly those codes. So this predicate alone does not pin self-reference; `Kleene.lean § I` states what that does and does not imply about codes generally, and `III` below gives the non-degenerate case.

Definition: <code>selfApply</code> and <code>IsComputationalQuine</code> (<code>Kleene.lean § I</code>)
<code>selfApply : Code → ℕ → . ℕ := fun c n ↦ eval c (encode c + n)</code>
A code c is a computational Quine if $\text{eval } c = \text{selfApply } c$, i.e.:
$\forall n, \text{eval } c \ n = \text{eval } c \ (\text{encode } c + n)$
This is a periodicity condition: c 's output at n equals its output at $\text{encode}(c) + n$. The encoding plays the role of the "address" of the program — c 's behavior at n is the same as c 's behavior at its own address plus n .
<code>selfApply_partrec</code> : <code>selfApply</code> is partially computable.
Proof: via <code>Mathlib</code> computability primitives — <code>eval</code> composed with <code>encode</code> and <code>nat_add</code> . Lean purity: standard foundational axioms only. ✓

Theorem: <code>computational_quine_exists</code>
There exists a computational Quine: $\exists c : \text{Code}, \text{IsComputationalQuine } c$.
Proof: immediate from <code>kleene_fixed_point_exists</code> applied to <code>selfApply</code> , using <code>selfApply_partrec</code> .
Lean purity: standard foundational axioms only. ✓
Note on uniqueness: the Quine atom of an <code>AFAStructure</code> lattice is unique by the class field <code>quine_unique</code> (<code>T-EXEC</code> , <code>ZP-J</code>). Codes meeting <code>IsComputationalQuine</code> are not unique: the constant codes alone give infinitely many (<code>infinite_quine_family</code>). Self-printing codes are unique at the level of behaviour (<code>selfprints_behaviour_injective</code> , <code>III</code> below). That the lattice's Quine atom is its $\perp$, and the only one, flows from <code>ZP-J T-EXEC</code> , not from either computational predicate.

III. Non-degenerate Self-Reference

A sharper predicate is not met by constants. A code is self-printing (`SelfPrints`) when, on every input of channel 0, it returns its own Gödel number, and universal (`Universal`) when, on channel $e + 1$, it computes what the code numbered e computes. A code that is both exists (`selfref_universal_exists`), and infinitely many do (`selfref_universal_infinite`). No constant code is universal (the control example after `selfref_universal_infinite` in `Kleene.lean § VIII`), and each half is met by some code without the other: `universal_not_selfprints`, and `Code.zero`, which is self-printing and not universal. These are statements about the partial function a code computes; they state that such codes exist, and none says that a code is run.

Abundance holds for every transformation: for any partially computable F , the codes c with $\text{eval } c = F(c)$ form an infinite set (`fixed_points_infinite`), and every partial recursive function has infinitely many codes (`padding`, the `Padding Lemma`). So the fixed points of self-application are many, and so are the codes of any one behaviour.

Theorem: `selfprints_behaviour_injective` (`Kleene.lean § VIII`)

For self-printing codes c_1 and c_2 : if $\text{eval } c_1 = \text{eval } c_2$ then $c_1 = c_2$.

Channel 0 reads each code's Gödel number back out of its behaviour, and the encoding is one-to-one.

Reading: computational self-reference has a uniqueness, at the level of behaviour. For self-printing codes this replaces the statement that computational quines are not unique; for the periodicity predicate `IsComputationalQuine` that statement stays true, since constant codes computing different functions all meet it. Both hold: infinitely many self-printing codes exist, and no behaviour belongs to two of them. Contrast `padding`, where infinitely many codes share one behaviour.

Lean: `ZeroParadox.selfprints_behaviour_injective`. Purity: `propext`, `Classical.choice`, `Quot.sound` (measured 2026-10-04); the predicate `SelfPrints` itself carries the same three.

Section II: KleeneStructure — Bridging AFA and Computation

I. The Bridge

ZP-J's `AFAStructure` typeclass encodes AFA self-containment in type theory: a predicate `selfMem`, AFA uniqueness (`quine_unique`), and the bridge field `bot_self_mem` ($\perp$ is self-containing). T-EXEC follows: the Quine atom equals $\perp$.

ZP-K's `KleeneStructure` extends `AFAStructure` with a computational witness: a code `botCode` satisfying the `selfApply` periodicity condition, whose existence is guaranteed by Kleene's theorem. The typeclass commitment takes AFA self-containment ($\perp = \{\perp\}$) and the Kleene fixed point to be the same structural role stated in two formal languages — this is the motivating claim, not a consequence derived by the theorems.

KleeneStructure Typeclass (`Kleene.lean § II`)

`class KleeneStructure (L : Type*) [ZPSemilattice L] extends AFAStructure L with:`

(inherited) `selfMem : L → Prop` — self-membership predicate

(inherited) `quine_unique` — AFA uniqueness

(inherited) `bot_self_mem` — $\perp$ is self-containing

(new) `botCode : Code` — a code selected from the existence proof; not a distinguished code

(new) `botCode_is_quine : IsComputationalQuine botCode` — a CLASS FIELD, not a theorem: the periodicity condition $\text{eval } c \ n = \text{eval } c$ (`encode c + n`), which constant codes also satisfy

KleeneStructure Typeclass (Kleene.lean § II)

(new) bot_self_mem_from_kleene : selfMem $\perp$ — a RESTATEMENT of the inherited AFAStructure field bot_self_mem, adding no content beyond it. Not an implication from the Kleene side.

Any KleeneStructure instance must supply both the AFA witness (bot_self_mem) and the computational witness (botCode with botCode_is_quine). Both are class FIELDS — assumptions discharged at instantiation, not results proved here. They are required together because the framework READS them as one structural fact; that reading is the motivation for the class, and it is not derived within it.

II. Why Not a Bridge Axiom?

The identification between AFA self-containment and Kleene computational fixed points is not asserted as a new axiom. KleeneStructure is a typeclass: any concrete type claiming this identification must discharge it as a proof obligation. The commitment is checked at instantiation, not accepted globally.

The distinction from ZP-J carries over: a freestanding axiom says "trust me." A typeclass field says "prove it for your specific type, or it does not compile." The MachinePhase instance in § V shows how the obligation is discharged concretely.

Section III: T-COMP — The Three-Way Equivalence, and the Bundled Computational Condition

ZP-J T-EXEC established a three-way equivalence: Quine atom (set-theoretic) $\leftrightarrow$ bottom element (order-theoretic) $\leftrightarrow$ join identity (algebraic). T-COMP proves exactly those three equivalent. ZP-K bundles a fourth face — Kleene fixed point (computational) — as a KleeneStructure class field, so all four are present simultaneously in any KleeneStructure lattice, but the fourth is required at instantiation rather than proved equivalent to the others.

Remark R-K.0 — What T-COMP Proves, and What "Four-Way" Meant

The equivalence among (1)–(4) has two distinct sources:

(1)–(3) are equivalent by T-EXEC (ZP-J): any element that is a Quine atom is also $\perp$ and a join identity, and vice versa. This is a genuine logical derivation — the three properties are proved to coincide from the AFAStructure axioms.

(4) is present in any KleeneStructure instance because KleeneStructure requires it as a typeclass field: botCode_is_quine must be supplied at instantiation. There is no independent proof that satisfying condition (1) (being a Quine atom in the AFA sense) entails satisfying condition (4) (being a Kleene fixed point), or vice versa. The two are combined by the typeclass definition — they are required together because we take them to be the same structural fact, not because one is derived from the other.

Remark R-K.0 — What T-COMP Proves, and What "Four-Way" Meant

Stated at the level of types, which is where the gap is visible: conditions (1)–(3) are properties of one element q of the lattice L , and T-COMP relates them to each other. Condition (4) is a property of a Code. There is no function, and no equivalence, between Code and L anywhere in the development — so the fourth condition is not connected to the other three by a mapping; it is required alongside them by the class. `da1_paths_unified` records exactly this and no more: it is a conjunction of a fact about `bot` in L and a fact about `botCode` in Code.

In short: "four-way equivalence" means "all four hold in any KleeneStructure instance." (1)–(3) are independently proved equivalent. (4) is bundled in by the typeclass requirement. The philosophical claim — that Kleene computational self-reference and AFA set-theoretic self-reference are the same thing — is the motivation for the typeclass design, not a consequence derived within it.

Theorem T-COMP — Computational Grounding (Kleene.lean § III)

In any KleeneStructure lattice L , for any $q : L$, the following are equivalent:

(1) `IsQuineAtom q` — set-theoretic self-reference (AFA)

(2) $q = \perp$ — order-theoretic minimum (ZP-A)

(3) $\forall x : L, \text{join } q \ x = x$ — algebraic generator (ZP-A A4)

Separately, the KleeneStructure requirement: $\exists \text{ botCode} : \text{Code}, \text{IsComputationalQuine botCode}$ (computational self-reference). It is not a clause of `t_comp`; it is present in any KleeneStructure instance because `botCode_is_quine` is a required field. The equivalence of (1)–(3) is derived by T-EXEC.

Lean: `ZeroParadox.t_comp`. Purity: standard foundational axioms — from Mathlib computability. ✓

I. Why Four Languages?

DA-1's three informal paths (Path 1: AFA structural, Path 2: informational, Path 3: Kolmogorov/computational) were previously understood as three separate corroborations converging on the same conclusion. In ZP-E each argues for DA-1's precondition, which is what the occurrence commitment asserts, and none derives it. Path 1's Lean counterpart is `da1_closed_concrete`, which proves `IsQuineAtom` ($\perp : \text{MachinePhase}$) and nothing computational; Path 3's Lean counterpart is the `machinePhaseKleene` instance's `botCode_is_quine` field, a KleeneStructure requirement that constant codes also meet, not a witness of execution and not a second independent proof.

Path 1 says: nothing external to $\perp$ can execute $\perp$, so if $\perp$ executes at all, the executor is $\perp$ itself, read as $\perp = \{\perp\}$; that rules out an external executor, not an inert $\perp$. Path 3 says: no shorter external program generates $\perp$; that rules out a shorter external generator, not an inert string, so executing is not derivable from incompressibility. The framework READS these as one claim in two vocabularies. What is proved is weaker and is a conjunction, not an identity: both hold together in any KleeneStructure. They cannot be equated formally — one is a statement about an element of the lattice, the other about a Code, and an equation across those types is not a well-formed proposition.

Theorem: da1_paths_unified (Kleene.lean § IV)

In any KleeneStructure lattice:

$\text{IsQuineAtom } \perp \wedge \text{IsComputationalQuine botCode}$

A CONJUNCTION: the AFA self-containment argument (Path 1) and the Kleene computational fixed-point argument (Path 3) both hold, simultaneously witnessed by the KleeneStructure instance. That they are the SAME fact is a framework reading, not a clause of this theorem — the second conjunct is the class field botCode_is_quine.

Lean: ZeroParadox.da1_paths_unified. Purity: standard foundational axioms only. ✓

II. The Description-Instantiation Gap

The "description-instantiation gap" asks whether mathematical self-reference implies computational execution: whether a self-referential $\perp$ is running (Sense B, ZP-E DA-1 insert § II) rather than an inert description (Sense A).

On the framework's reading, $\perp$ in the computational instantiation is read as the universal Turing machine in its ground state, which is not a description awaiting an external executor: if it executes at all, it executes itself (Path 1, the framework's requirement, not a theorem). That rules out an external executor, not an inert $\perp$: on MachinePhase, $\perp$ is the Quine atom (da1_closed_concrete), and T-SNAP holds in a dynamics in which nothing leaves c_0 (tsnap_holds_but_nothing_moves). That it does execute, so that the configuration reaching P_0 is running, is the occurrence commitment, which DA-1 consumes and does not supply (ZP-E).

Theorem: description_instantiation_gap_closed (Kleene.lean § IV)

In any KleeneStructure lattice:

$\text{IsQuineAtom } \perp \wedge \forall q : L, \text{IsQuineAtom } q \rightarrow q = \perp$

On the framework's reading, $\perp$ is not a description awaiting an external interpreter: if it executes, it is its own executor, the universal machine in its ground state. Lean proves only the Quine-atom statement above; that $\perp$ executes is the occurrence commitment, not this theorem. Lean witnesses $\perp$ as the AFA Quine atom of MachinePhase (da1_closed_concrete) and carries the Kleene quine as a KleeneStructure requirement (botCode_is_quine); that these are one structural fact is the framework's reading.

Lean: ZeroParadox.description_instantiation_gap_closed. Purity: standard foundational axioms only. ✓

A description has two uses. Cotler, Hongler and Hudcová (arXiv:2510.08342, 2025, p. 2) report von Neumann's self-replicating cellular automaton (their ref. [1]) as "leveraging the dual use of an organism's description – once to be interpreted for construction and once to be transcribed to the new offspring". Reading: the transcribed use is Sense A, a description copied as data, and the interpreted use is Sense B, a description being run (ZP-E, DA-1 insert § II). A description present on a tape supplies the first use and not the second.

Universality does not give replication. The same authors construct a "non-talking heads" cellular automaton that is locally Turing-universal and yet cannot sustain non-trivial self-replication (their Theorem 2.8, p. 8): "any head encountering a cell marked by another head halts immediately. This blocking of information transfer prevents self-replication" (p. 7). Their Theorem 2.9 (p. 8) places self-replication strictly between two

strengths of universality, $\text{GloballyUniversal} \subsetneq \text{UniversalSelfReplicating} \subsetneq \text{LocallyUniversal}$, and they note that "a quine produces a description of the copying mechanism without replicating the mechanism itself" (p. 8). Reading: the self-printing universal codes of Section I.III are quines in that sense. Each returns a description of itself, its own Gödel number, on channel 0: the description in Sense A.

Occurrence is not read off from outside. `carry_steps_onward_forever_yet_shows_nothing` (Occurrence.lean § VI-c): a deterministic step that, at every state, moves to a different state and never halts, while its observable stays constant on everything reachable. Reading: an observer of that observable sees the same constant trace as from a dynamics in which nothing moves (`stuckPhase`, `tsnap_holds_but_nothing_moves`), so whether a configuration is running cannot be settled by watching what it shows.

Reading, in two charts, neither denied. History chart: a complete description of a run, every configuration in order, IS the run; nothing further is left to happen. State chart: a description, even a complete one, is a configuration, and that it is running is a further fact, the occurrence commitment, which DA-1 consumes and does not supply (ZP-E). Both are readings.

Section IV: Axiom Footprint

This section records measurements, not a rule. `#print axioms` traverses a declaration's statement as well as its proof term, and reports what a proof used — never what a theorem needs. The following were measured on 2026-09-19 against the pinned Mathlib; re-run them rather than citing this table if the answer matters.

Declaration	Axiom footprint
<code>Nat.Partrec.Code</code>	none
<code>Nat.Partrec.Code.eval</code>	none
<code>IsKleeneFixedPoint</code>	none
<code>encodeCode_self(encodeCode c = encodeCode c, by rfl)</code>	none
<code>encode_self(Encodable.encode c = Encodable.encode c, by rfl)</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>kleene_fixed_point_exists</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>roger_fixed_point_exists</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>machinePhaseKleene</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>da1_closed_concrete</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>machinePhaseAFA</code>	none
<code>bot_is_quine_atom</code>	none
<code>t_exec</code>	none

Two readings of that table are worth naming, and neither generalises beyond the rows above. The two spellings of a code's Gödel number differ: the raw `encodeCode` is axiom-free, while `Encodable.encode` reaches through Mathlib's `Denumerable Code` instance, whose own term spends `Classical.choice`. And one

statement is proved both ways in Kleene.lean §V — IsQuineAtom (bot : MachinePhase) via machinePhaseKleene carries the triple, and the same statement via machinePhaseAFA carries nothing. The axioms do not enter through ZPSemilattice or AFAStructure.

The Kleene.lean § VIII and Disjunctive.lean declarations cited in Section I.III and Section VI are not in that table. Their #print axioms lines are in those two files' PurityCheck sections, and where Classical.choice enters selfref_universal_exists is recorded in ZeroParadox/Computability/Kleene.md § VIII, measured 2026-10-04.

ZP-J T-EXEC and all its corollaries remain axiom-free. The classical axioms are entirely localised to the computational layer.

Remark: Classical Choice in Computability

Where the measurements above show Classical.choice, it is reached through Mathlib's numbering of program codes (Denumerable Code): it is present even in a computable instance with a constant code. ZP-K's machinePhaseKleene also picks botCode with Classical.choose, which is what makes that instance noncomputable. Whether the numbering's footprint is essential is not measured. Kleene's theorem is an existence result, and the code it supplies is selected non-constructively. This is standard in computability theory — the theorem guarantees existence without giving a canonical construction.

The MachinePhase instance (§ V) uses Classical.choose to pick botCode from the existence proof. This makes machinePhaseKleene noncomputable, which is correct and expected.

Section V: MachinePhase Instance — DA-1: what Lean witnesses

I. The Concrete Instantiation

ZP-E's MachinePhase type is the two-element type {initial, running} carrying the ZPSemilattice instance (bot = initial = c_0 , join = binary maximum). ZP-J gave it AFAStructure via the selfMem predicate. ZP-K gives it KleeneStructure by adding the computational witness botCode.

The selfMem definition for MachinePhase is: selfMem $x := x = \perp$. This is the CIC-compatible encoding of AFA self-containment: "self-containing" means "equals the bottom element." Anti-foundation is not required at the typeclass level — the relevant structural fact ($\perp$ is the unique self-containing element) is captured by the definition and proved by rfl.

AFAStructure MachinePhase Instance (Kleene.lean § V)

instance machinePhaseAFA : AFAStructure MachinePhase where

selfMem $x := x = \perp$ (self-containing = equals initial state)

quine_unique _ _ hx hy := hx.trans hy.symm (if $x = \perp$ and $y = \perp$ then $x = y$)

bot_self_mem := rfl ($\perp = \perp$, proved by reflexivity)

AFAStructure MachinePhase Instance (Kleene.lean § V)

This is the CIC encoding of $\perp = \{\perp\}$: the initial machine state is self-containing and is the unique element with this property. No ZF+AFA axiom is added — the structural fact is encoded as a definition that Lean verifies.

KleeneStructure MachinePhase Instance (Kleene.lean § V)

noncomputable instance machinePhaseKleene : KleeneStructure MachinePhase where

botCode := Classical.choose computational_quine_exists

botCode_is_quine := Classical.choose_spec computational_quine_exists

bot_self_mem_from_kleene := rfl

botCode is selected non-constructively from the existence proof provided by Kleene's theorem — an arbitrary witness, not a distinguished code. Reading it as "the program that IS its own program" is the framework's commitment carried by the class, not a property this instance establishes: IsComputationalQuine is a periodicity condition, strictly weaker than self-reference.

II. DA-1: what Lean witnesses

With the MachinePhase KleeneStructure instance in place, the abstract theorem da1_computational (which holds for any KleeneStructure lattice) applies directly to ZP-E's machine. The result is concrete.

Theorem da1_closed_concrete — Path 1's Lean counterpart (Kleene.lean § V)

da1_closed_concrete : IsQuineAtom ($\perp$: MachinePhase)

The initial machine state c_0 is a Quine atom: it is self-containing and is the unique self-containing element of the MachinePhase lattice.

Interpretation (the framework's reading, not a Lean theorem): c_0 is read as the universal Turing machine in its ground state, needing no external executor: if it executes at all, it executes itself (Path 1). That it executes is the occurrence commitment, which DA-1 consumes, not a consequence of this theorem.

Lean: ZeroParadox.da1_closed_concrete. Purity: standard foundational axioms only (the same statement has an axiom-free proof, bot_is_quine_atom). ✓

III. What Changed for DA-1

ZP-E's DA-1 section previously carried the designation "Outside Lean Scope" with three justifications: Path 1 requires ZF+AFA (incompatible with Lean's CIC/MLTT); Path 3 requires Kolmogorov complexity (uncomputable, absent from Mathlib); Path 2 requires a step from unbounded surprisal to executing, which ZP-E names a bridge principle of its own, a missing principle and not a missing proof.

For Path 1 (AFA structural), the Lean counterpart is `da1_closed_concrete`, which proves `IsQuineAtom (⊥ : MachinePhase)` and nothing computational: in the `AFAStructure` typeclass, `selfMem` encodes $\perp = \{\perp\}$ in CIC-compatible form, and the proof obligation is discharged by the `MachinePhase` instance. Path 3 (computational) is NOT resolved here. Its Lean counterpart, `botCode_is_quine`, is a `KleeneStructure` class field — an assumption supplied at instantiation, not a second independent proof — and the condition it requires, `IsComputationalQuine`, is a periodicity condition that a constant code satisfies trivially. The Kolmogorov reading ("no shorter program is prior to $\perp$ ") has no formal content in ZP-K: Kolmogorov complexity is uncomputable and absent from the development. Path 3 therefore derives nothing here: in ZP-E it bears on DA-1's precondition and shows that executing is not derivable from incompressibility, and that precondition is what the occurrence commitment asserts.

Path 2 (informational: unbounded surprisal $\rightarrow$ executing) is a bridge principle of its own — a missing principle, not a missing proof — which the framework does not adopt, which is not the occurrence commitment, and which is not a premise of the Snap. The mathematics of L-INF ($\perp_{\text{inf}}$) is proved; but the step from "exceeds every finite informational bound" to "therefore executing" asks what it means for a mathematical structure to instantiate rather than merely satisfy conditions. No computability library answers this question. Importantly, DA-1 does not depend on Path 2: the formal spine (`DP-2 + da1_minimal_path`) is proved axiom-free. Path 2 is motivational context.

DA-1 Lean scope status after ZP-K: Path 1 (structural, AFA): Lean counterpart `da1_closed_concrete : IsQuineAtom (⊥ : MachinePhase)`, nothing computational; it rules out an external executor, not an inert $\perp$. Path 3 (computational, Kleene): `botCode_is_quine` is a `KleeneStructure` class field, assumed at instantiation; Path 3 shows executing is not derivable from incompressibility. Path 2 (informational, L-INF): UNADOPTED BRIDGE PRINCIPLE — a missing principle, not a missing proof, and not the occurrence commitment. None derives DA-1's precondition, which is what the occurrence commitment asserts.

Section VI: Replication — Presence, Address, and Selection

This section follows one arc, one declaration per step, from the presence of every instruction on a tape to the selection of one replica. Each step is a statement about a tape $\mathbb{N} \rightarrow \text{Bool}$ or about the partial function a code computes. Presence on a tape is Sense A (ZP-E, DA-1 insert § II); that anything runs is the occurrence commitment, as in Section III.II.

I. Presence: Every Instruction Is on the Tape

A tape is disjunctive (the standard term; also "rich") when every finite word occurs in it infinitely often (Disjunctive; Barnsley and Leśniak, arXiv:1203.0481, § 3). On a disjunctive tape the binary of every program code occurs infinitely often (`code_occurs_of_disjunctive`). That is presence of the description, Sense A; nothing in the statement reads or runs a code. Disjunctive does not need randomness: `champ`, a Champernowne-style tape, is disjunctive and primitive recursive (`champ_disjunctive`, `champ_primrec`), and almost every fair-coin tape is disjunctive (`fairTape_disjunctive_ae`).

Reading: under the framework's commitment that $\perp$ is maximally complex, $\perp$ read as a tape $\mathbb{N} \rightarrow \text{Bool}$ would have incompressible prefixes, hence be Martin-Löf random (Levin–Schnorr), and Martin-Löf random sequences are disjunctive in standard theory, so every instruction would be present. That implication is not proved in Lean: Martin-Löf randomness and Kolmogorov complexity are not located in the Mathlib pin as of

2026-10-04 (searches recorded in ZeroParadox/Information/Disjunctive.md). The converse fails: champ is disjunctive and computable.

II. Replicas Differ Only by Address

Theorem: selfPrints_universal_address (Kleene.lean § VIII)

For self-printing universal codes c and d : they agree on every interpreter channel $e + 1$, and if $c \neq d$ they differ on channel 0, where each returns its own Gödel number.

The control beside it in Kleene.lean shows that Universal carries the first half: Code.zero is self-printing and not universal, and it disagrees with a self-printing universal code on a channel $e + 1$.

Reading: replicas share everything but the address. Incompressibility is a property of each instance, and the next instance adds only its address. That is a claim about relative complexity, and it stays a reading while Kolmogorov complexity is not in Lean.

Lean: ZeroParadox.selfPrints_universal_address. Purity: propext, Classical.choice, Quot.sound (measured 2026-10-04).

III. No Full Self-Copy on a Disjunctive Tape

A disjunctive tape equals its own shift by no offset $a > 0$ (disjunctive_not_periodic; not_disjunctive_of_periodic is the same fact read from the periodic side). At offset 0 every tape equals its own shift, champ included (the control example beside it in Disjunctive.lean § V). Reading: a tape carrying every code word holds no full copy of itself at any positive offset; copies sit side by side, told apart by an address (II).

IV. Constructing an Address, and Selecting One

Constructing an address is the recursion theorem's fixed point of a computable transformation: selfref_universal_exists takes its code from Mathlib's fixed_point₂ applied to a partially computable map (selfPrintOrDelegate_partrec). Selecting a code by its behaviour is selection by a predicate that is not decidable: membership in IsComputationalQuine is not a ComputablePred (isComputationalQuine_undecidable), and nor is membership in any non-trivial extensional set of codes (the second conjunct of quine_exists_yet_rice, Rice's theorem). ChoiceCannotBe.lean locates where choice does its work: when the predicate is decidable, selection is free, computed by if (select_of_decidable), while the uniform selection principle at an undetermined pole is the choice fragment, definitionally (uniformChartSelection_iff_choiceFragment).

Reading: constructing an address is a computation, and selecting one by behaviour is where choice works. The Lean footprint does not draw that line, and is stated as measured: selfref_universal_exists carries Classical.choice, and so does its statement, through Mathlib's numbering of codes (Denumerable Code, reached through ofNatCode); restated with encodeCode, the statement is axiom-free, and no proof of the restated form without choice was found. ZeroParadox/Computability/Kleene.md § VIII records the measurement (2026-10-04) and what that route lacked as located then: a recursion theorem stated without Partrec₂, and a universal code proved correct without eval_part. The footprint is UNCLASSIFIED.

V. The Zero Tape: Reference and Self

In $\mathbb{N} \rightarrow \text{Bool}$ under the pointwise order, $\perp$ is the all-false tape, and that tape is not disjunctive (`allFalse_not_disjunctive`); it carries the word of no code other than `Code.zero`, whose word is empty (`allFalse_misses_code`, `codeWord_zero_width`). Under pointwise exclusive-or, every tape's self-difference is the all-false tape, and read against the all-false tape a tape is itself (the examples in `BottomCannotBe.lean`, "The infinitude in two charts: content and self").

Reading, in two charts, neither denied: the all-false tape, $\perp$ of the pointwise tape order, is the reference every comparison runs through (content chart) and every tape's self-difference (self chart), one object in two charts. That the framework's $\perp$ is maximally complex is a commitment, stated as no equation with any one tape.

VI. What Replication Needs Beyond Self-Reference

In Cotler, Hongler and Hudcová's setting, replication needs information to cross a boundary, "inter-head communication is crucial for replication" (p. 8): their non-talking heads automaton blocks that crossing and is locally universal without self-replication (Theorem 2.8, p. 8). A second capacity separates the two strengths of universality, `GloballyUniversal` $\subsetneq$ `LocallyUniversal` (their Theorem 2.3, p. 5): of the reversible automata, "Some can locally implement reversible universal Turing machines but cannot globally simulate irreversible CAs". Reading: these are two different overwrite capacities, writing into a cell another head has marked, which replication needs, and irreversible erasure, which such a reversible automaton cannot simulate globally. This mapping onto the framework is a reading.

Traceability Register — ZP-K

Claim	Grounded In	Axioms	Status
<code>selfApply_partrec</code>	<code>eval_part</code> (Mathlib) + <code>Primrec.encode</code> + <code>Primrec.nat_add</code>	Standard Mathlib foundational axioms	Lean: <code>selfApply_partrec</code> ✓
<code>computational_quine_exists</code>	<code>kleene_fixed_point_exists</code> + <code>selfApply_partrec</code>	Standard Mathlib foundational axioms	Lean: <code>computational_quine_exists</code> ✓
T-COMP: three-way equivalence	ZP-J T-EXEC (<code>t_exec_triple_iff</code>)	Standard Mathlib foundational axioms	Lean: <code>t_comp</code> ✓
<code>da1_paths_unified</code>	<code>bot_is_quine_atom</code> + <code>botCode_is_quine</code>	Standard Mathlib foundational axioms	Lean: <code>da1_paths_unified</code> ✓
<code>description_instantiation_gap_closed</code>	<code>bot_is_quine_atom</code> + ZP-J <code>t_exec</code>	Standard Mathlib foundational axioms	Lean: <code>description_instantiation_gap_closed</code> ✓
<code>machinePhaseAFA</code> (<code>AFAStructure</code>)	<code>selfMem := x = ⊥</code> ; <code>quine_unique</code> ; <code>bot_self_mem</code> <code>:= rfl</code>	No axioms	CIC encoding of $\perp = \{\perp\}$ for <code>MachinePhase</code> ✓

Claim	Grounded In	Axioms	Status
machinePhaseKleene (KleeneStructure)	machinePhaseAFA + Classical.choose computational_quine_exists	Standard Mathlib foundational axioms	noncomputable — classical choice for botCode ✓
da1_closed_concrete	da1_computational + machinePhaseKleene	Standard Mathlib foundational axioms	Lean: da1_closed_concrete ✓ DA-1 structural half — no Code, no execution (definitionally: under selfMem $x := x = \perp$, reduces to $(\perp = \perp) \wedge (\forall x, x = \perp \Rightarrow x = \perp)$; structural closure by typeclass design — see R-K.0)
fixed_points_infinite, padding	fixed_point ₂ _unbounded (Kleene.lean § VIII)	propext, Classical.choice, Quot.sound (measured 2026-10-04)	Lean ✓ — infinitely many fixed points for every partially computable F; the Padding Lemma
selfref_universal_exists, selfref_universal_infinite	fixed_point ₂ (Mathlib) + selfPrintOrDelegate_partrec; fixed_points_infinite	propext, Classical.choice, Quot.sound (measured 2026-10-04); UNCLASSIFIED, Kleene.md § VIII	Lean ✓ — existence of self-printing universal codes; nothing says a code is run
selfprints_behaviour_injective	SelfPrints at channel 0 + Encodable.encode_injective	propext, Classical.choice, Quot.sound (measured 2026-10-04)	Lean ✓ — Reading: a uniqueness at the level of behaviour
selfPrints_universal_address	SelfPrints + Universal + Encodable.encode_injective	propext, Classical.choice, Quot.sound (measured 2026-10-04)	Lean ✓ — Reading: replicas share everything but the address
code_occurs_of_disjunctive	Disjunctive (Disjunctive.lean § I) at each code word	propext, Classical.choice, Quot.sound (measured 2026-10-04)	Lean ✓ — presence, Sense A; no code is read or run
champ_disjunctiveOnce, champ_primrec	tri / untri slot layout + bitAt_ofBits; Mathlib Primrec	champ_disjunctiveOnce: propext, Quot.sound; champ_primrec: propext, Classical.choice, Quot.sound (measured 2026-10-04)	Lean ✓ — disjunctive without randomness
fairTape_disjunctive_ae	second Borel–Cantelli (measure_limsup_eq_one) on disjoint aligned blocks	propext, Classical.choice, Quot.sound (measured 2026-10-04)	Lean ✓ — almost every fair-coin tape is disjunctive
disjunctive_not_periodic	not_disjunctive_of_periodic (Disjunctive.lean § V)	propext, Classical.choice, Quot.sound (measured 2026-10-04)	Lean ✓ — no full self-copy at any offset $a > 0$

Claim	Grounded In	Axioms	Status
carry_steps_onward_forever_yet_shows_nothing	stutter_obs_const + carry_stutters (Occurrence.lean § VI-c)	propext, Classical.choice, Quot.sound (measured 2026-10-04)	Lean ✓ — steps onward forever with a constant observable
Reading: zero tape (Section VI.V)	BottomCannotBe.lean, "The infinitude in two charts: content and self"	— (interpretation)	Reading, not a theorem: the all-false tape as reference and as self-difference
Reading: complete description, two charts (Section III.II)	occurrence commitment (ZP-E)	— (interpretation)	Reading, not a theorem: history chart and state chart, both stated

Open Items Register — ZP-K

Item	Status	Description
DA-1 Path 1 (AFA structural)	Lean counterpart proved; derives no execution	IsQuineAtom ($\perp$: MachinePhase) — the structural half, Path 1's Lean counterpart. The theorem mentions no Code and no execution. Path 1 says that if $\perp$ executes, it executes itself (the framework's requirement, not a theorem): it rules out an external executor, not an inert $\perp$. That $\perp$ executes is the occurrence commitment, which DA-1 consumes; Path 1 does not close it.
DA-1 Path 3 (computational)	ARGUES FOR THE PRECONDITION — derives nothing	botCode_is_quine is a KleeneStructure class field — assumed at instantiation, not proved. IsComputationalQuine is a periodicity condition satisfied by constant codes, and no Kolmogorov content exists in ZP-K. In ZP-E, Path 3 shows that executing is not derivable from incompressibility; DA-1's precondition is what the occurrence commitment asserts.
DA-1 Path 2 (informational)	UNADOPTED BRIDGE PRINCIPLE	L-INF (l_{inf}) is proved. The bridge "unbounded surprisal $\rightarrow$ executing" is a bridge principle of its own, a missing principle, not a missing proof; the framework does not adopt it, and it is not the occurrence commitment or a premise of the Snap. The gap between 'system at P_0 ' and 'system is running' cannot be closed by any computability library. DA-1 does not depend on Path 2.
selfApply uniqueness	CLOSED — not attempted (correct)	Codes meeting IsComputationalQuine are not unique: the constant codes alone give infinitely many (infinite_quine_family). That the Quine atom is $\perp$ of the lattice, and the only one, flows from ZP-J T-EXEC (set-theoretic side). For self-printing codes, a uniqueness at the level of behaviour is proved (selfprints_behaviour_injective).
Rogers' fixed-point theorem	CLOSED — roger_fixed_point_exists	For any computable $f : \text{Code} \rightarrow \text{Code}$, $\exists c$, $\text{eval}(f\ c) = \text{eval}\ c$. Lean: roger_fixed_point_exists — standard foundational axioms only. ✓

Item	Status	Description
ZP-B MachinePhase instance	OPEN — future work	The 2-adic model from ZP-B (Q_2 structure) has not been given a KleeneStructure instance. This is a natural extension but not required for T-SNAP or DA-1.
Choice-free recursion theorem	OPEN — measured negative, 2026-10-04	The statement of selfref_universal_exists carries Classical.choice through Mathlib's numbering of codes (Denumerable Code); restated with encodeCode it is axiom-free, and no proof of the restated form without choice was found. As located then, the route lacks a recursion theorem stated without Partrec ₂ and a universal code proved correct without eval_part (Kleene.md § VIII). Footprint UNCLASSIFIED.
Martin-Löf randomness $\Rightarrow$ disjunctive	OPEN — not in Lean	Standard theory: a Martin-Löf random sequence is disjunctive. Martin-Löf randomness and Kolmogorov complexity are not located in the Mathlib pin as of 2026-10-04 (search record in Disjunctive.md). Until this bridge is formalised, the framework's commitment that $\perp$ is maximally complex reaches Disjunctive (Section VI.I) only as a reading.

End of ZP-K | Computational Grounding of Self-Reference | DA-1 structural half: da1_closed_concrete : IsQuineAtom ($\perp$: MachinePhase) - no code, no execution | Four-way equivalence (R-K.0): Quine atom, $\perp$ and join identity proved to coincide; the Kleene fixed point is required by the typeclass, not derived | Path 2: an unadopted bridge principle, not a missing proof | All Kleene.lean theorems verified. Axioms: standard Mathlib foundational axioms.